



Optimizing Revenue Leakage & Profitability in the Hospitality Sector

Executive summary

- The given report examines the revenue flow of the “Atliq Group of Hotels” across Delhi, Mumbai, Hyderabad, and Bangalore.
- Services are spread across 7 types of property, offering 4 types of rooms.
- The study was conducted over a period of three months, from May to July, in 2022.
- There were a total of 1,34,590 rooms booked out of 40,74,011.
- 24.83% of bookings were cancelled, and 70.15% were a success.
- The average stay duration of guests was 2 days. And, the average lead time for advance bookings was 4 days.

Executive summary

- Only 2.32% rooms were occupied.
- Weekends had less revenue loss in comparison to weekdays.
- The presidential room of Atliq Grands in Delhi had the least revenue loss during the period of time.
- Atliq Exotica in Mumbai had a major revenue loss during this time, out of all.
- Booking trends showed a surplus of bookings from “other” platforms.
- The cancellation rate was the lowest when the bookings were made through “LogTrip”.
- During the end of month and start of a new month, the revenue loss tends to decrease by average.



Introduction

In the competitive and dynamic hospitality industry, hotels and resorts often face unexplained revenue leakages, fluctuating profitability, and underutilized service offerings. These issues can stem from a variety of factors, such as inefficient pricing strategies, low performing service segments (e.g., spa, room service, event hosting), poor upselling or cross-selling, misaligned customer targeting, or seasonal demand mismanagement.

Objectives

WHY IT IS HAPPENING?

WHAT IS HAPPENING?

HOW TO IMPROVE IT?

Identifying the root cause
and operational analysis to
determine the factors for
substandard performance

Data preparation, data
exploration, surface level
analysis and gaining
insights on the trends

Proposal of new initiatives,
revenue and loss forecast,
improving the factors affecting
key performance indicators



- Used the dataset having information about the booking information of “Atliq Group of Hotels” during the months of May to July, 2022.
- The dataset consisted of multiple sheets having data on revenue generated, revenue realized, number of guests, details about the property, feedback from the visitors, etc.
- Individual CSV sheets were co-joined to form a big data file.

Data sources and Preprocessing



- 77907 rows where null in ratings columns.
- As more than 50% of values were missing, a semi-supervised imputation technique was employed to fill in the missing values.
- Imperative Imputer was used after one-hot encoding the dataset.
- All the data was saved in place.
- A new document was made consisting of new initiatives, the capital required and its affect on revenue geenration.

Data sources and Preprocessing



- We finally have the big data file and a new initiative file with all the proposed changes.
- The big data file has 20 columns with 1,34,590 entries.
- The new initiative's proposal file has 14 new proposals with their feasibility score based on historical data.
- This raw data was used for an elementary analysis using matplotlib and seaborn.
- All this data was imported into Tableau for deeper analysis.

Data sources and Preprocessing

Key Performance Indicators

KPI's for the overall dataset are given in the corresponding slides. More specific KPI's for different situations can be analyzed through Dashboard 1 in the attached Tableau workbook.



2.32%

OF THE TOTAL CAPACITY
WAS OCCUPIED FOR THE
GIVEN PERIOD OF TIME



₹492.77

WAS THE GENERATED
REVENUE PER AVAILABLE
ROOM



₹119.55

WAS THE GENERATED LOSS
PER OCCUPIED ROOM
(LOSS-POR)



3.61

WAS THE MEAN RATING
GIVEN BEFORE IMPUTATION

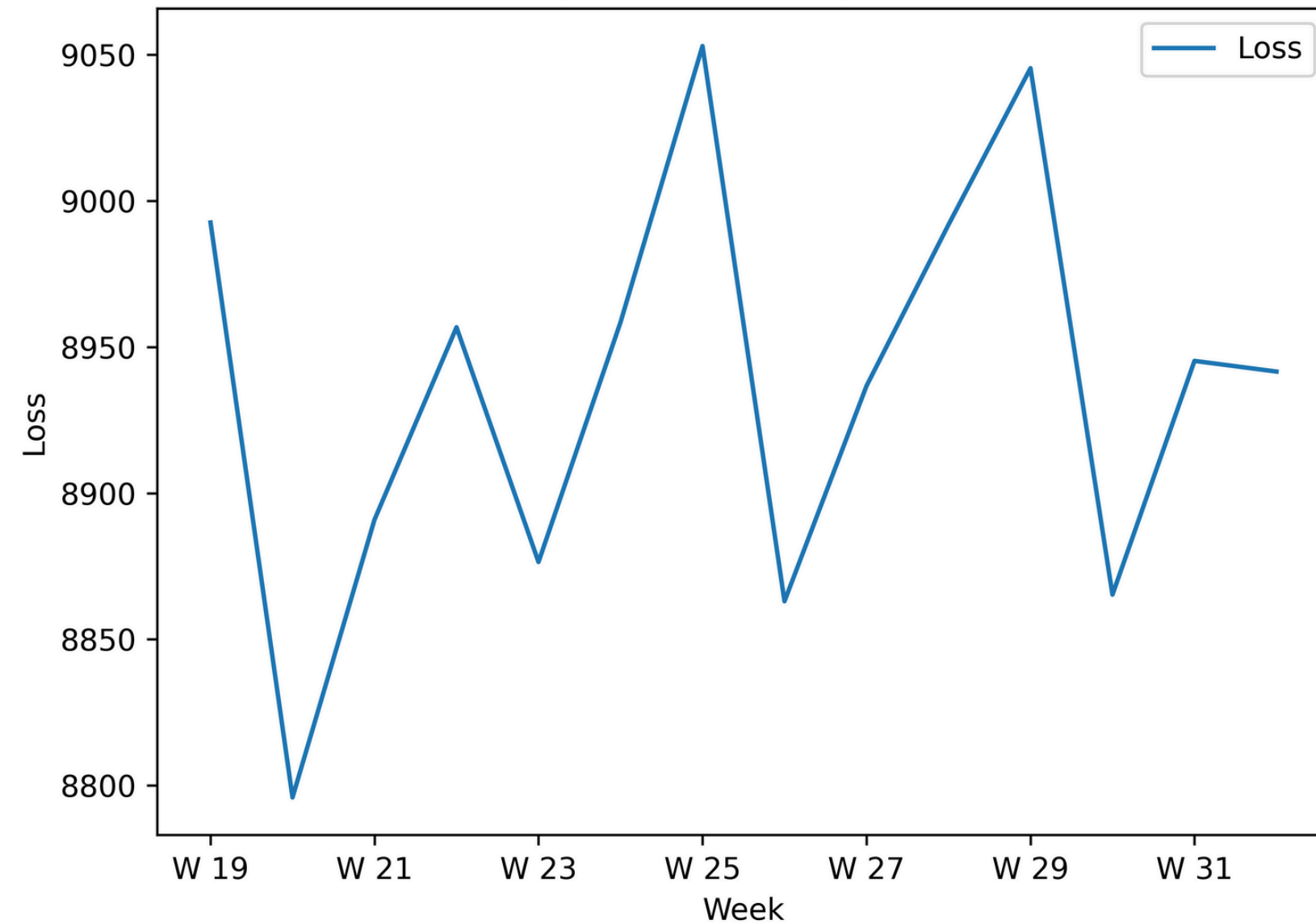
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WAS THE MEAN RATING
GIVEN AFTER IMPUTATION

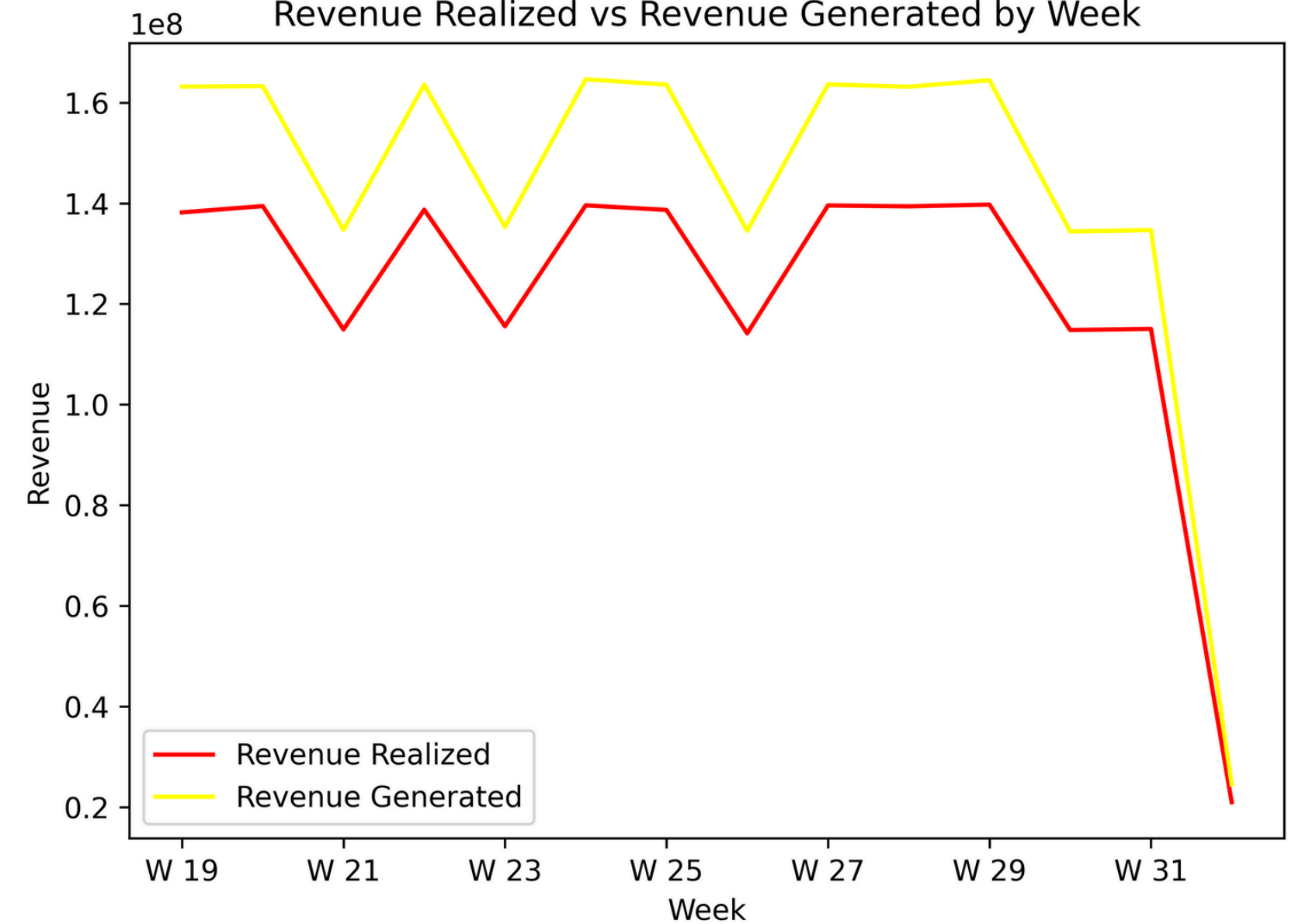


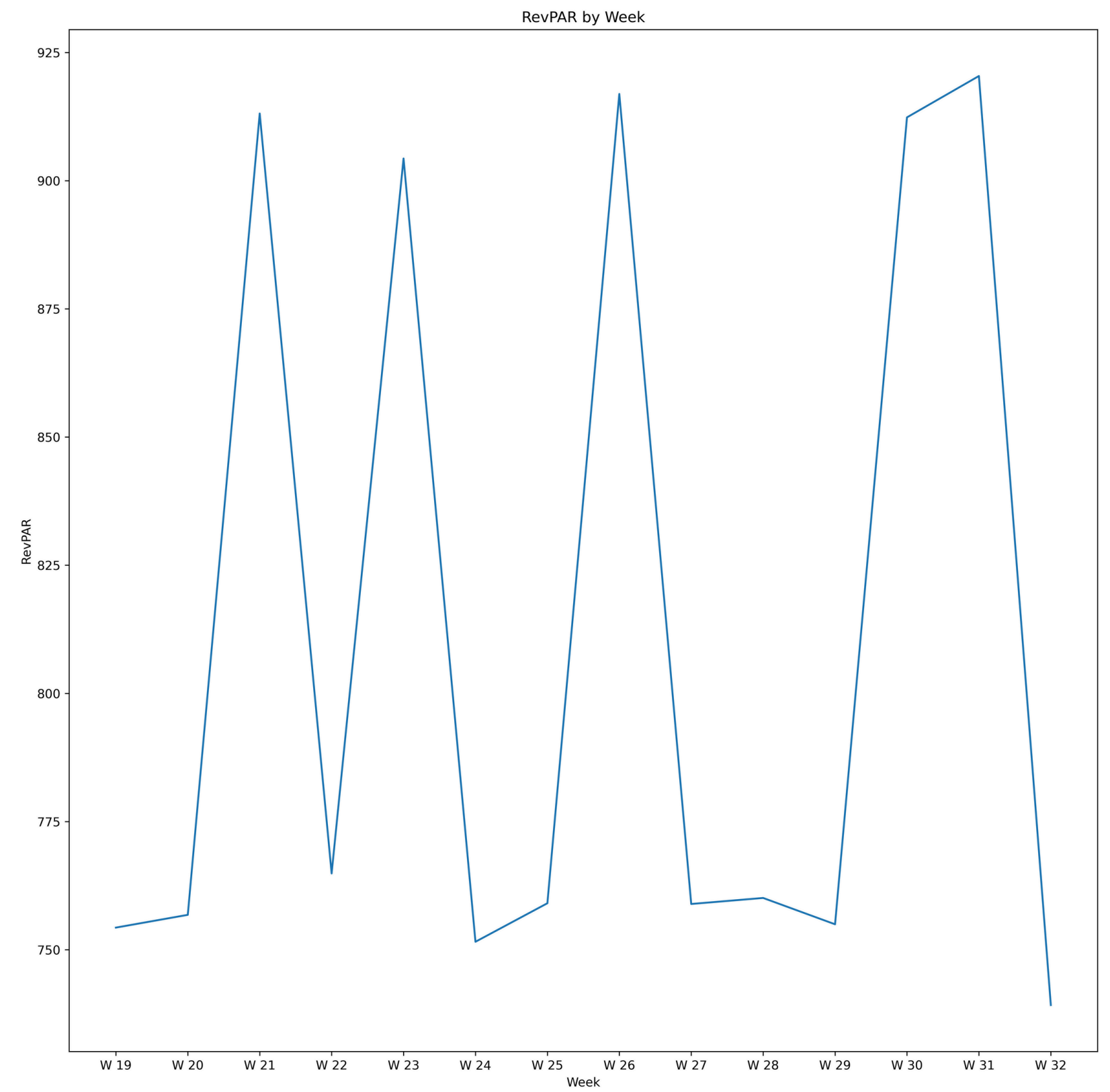
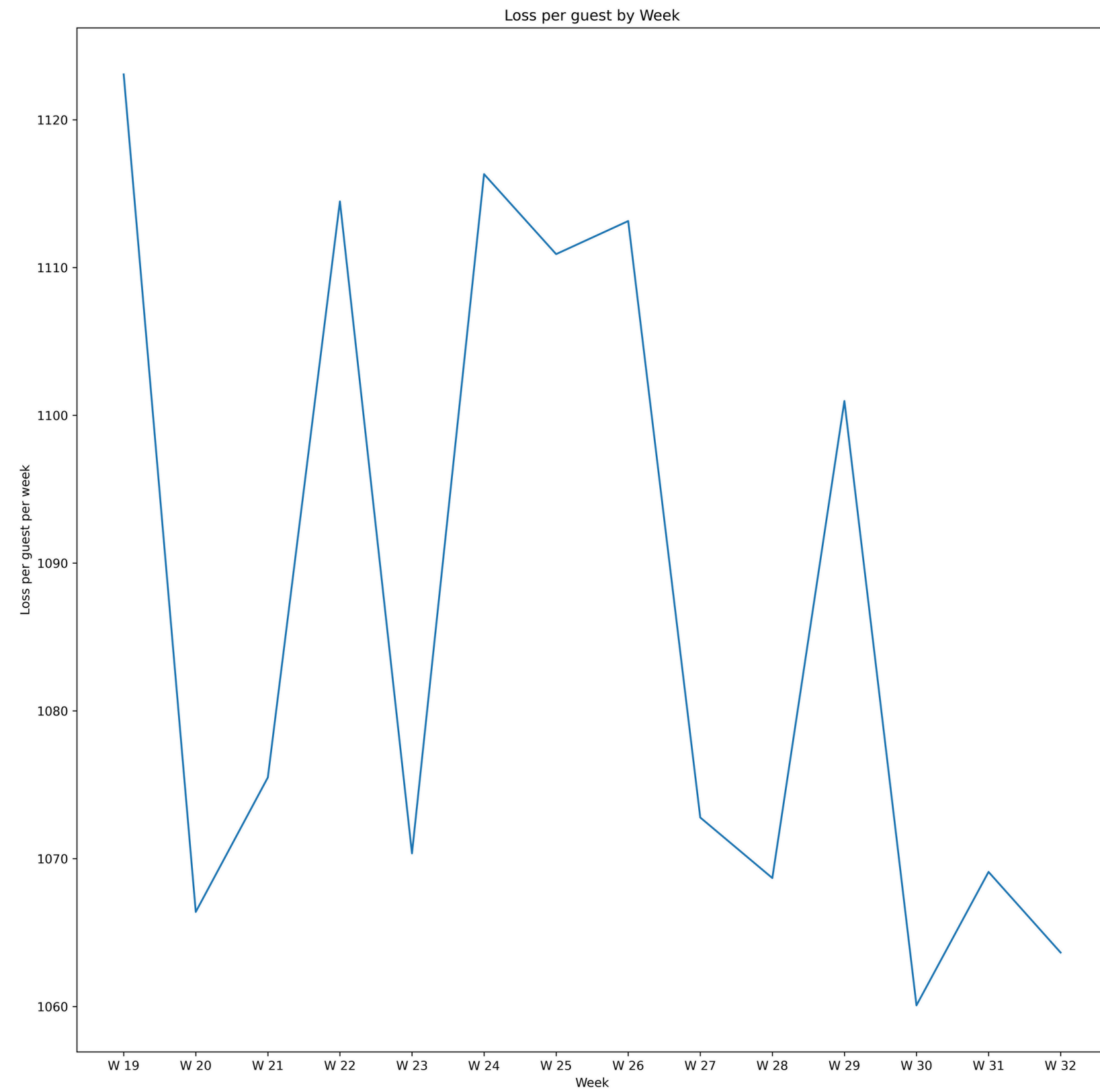
Current Performace Overview

Comparison of Loss per week



Revenue Realized vs Revenue Generated by Week

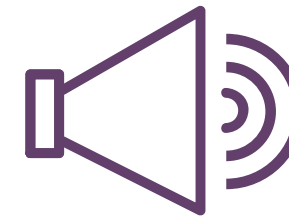




Root cause and operational analysis



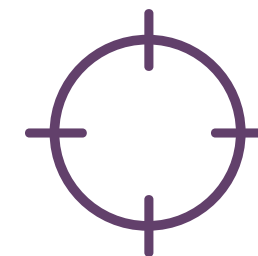
Occupancy percentage is similar for all room types.



RevPAR varies a lot with each room type.



Major revenue leakage is observed in the elite room type.



Longer stays had lesser cancellation rate.



Some services offered need to be normalised across all room types to decrease the revenue disparity.

Proposed changes



Price changes with respect to price elasticity. This can be easily visualised through Dashboard 3.

Increasing the expenditure on marketing. Exploring marketing avenues have tend to increase occupancy percentage.

Implementing new policies and bringing in essential changes. Based on historical data and feedback, this would decrease the revenue leakage.

Conclusion & Key Takeaways

-  Revenue Leakage is Concentrated → Losses are driven by high cancellation/no-show rates in specific cities and room types - targeted policy changes can reduce this by over 20%.
-  High-Impact Trends Identified → Clear seasonality and platform-wise booking behavior observed. Weekends and festive months show strong potential for bundled pricing.
-  Root Causes Are Actionable → Underperformance stems from weak booking policies, inefficient marketing allocation, and low weekday occupancy, not just pricing.
-  Data-Driven Initiatives Proposed → Each recommendation is backed by financial modeling, with feasibility scores and ROI metrics to guide decision-making.
-  Strategic Investments Show Strong Payback → Several initiatives (like dynamic pricing, loyalty programs, and couple/corporate packages) offer payback within 3-6 months.
-  Path to Profitability is Clear → By implementing just the top 3-4 feasible ideas, properties can unlock ₹1-2L/month in incremental revenue across key regions.

Use the attached tableau workbook for more self-explanatory analysis.

The slides are exhaustive, covering all of the important topics. But going through the other files is necessary.

This is a data-backed analysis. We may extend it to other scenarios. But, having a proper analysis of the data before reaching to any destination is very important.



Important!

Thank you

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